Solution Requirements

Date	15 March 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users securely log in using a username/email and password.	High
2	Authorization levels	Admin can manage the system and models; users can submit applications and view prediction results.	High
3	External interfaces	The system integrates with the Flask web application and IBM Watson Machine Learning for model deployment.	High
4	Transactions processing	The system accepts applicant details, processes them using the ML model, and returns an approval or rejection prediction.	High

5	Reporting	Generate prediction results and maintain applicant records for future reference.	Medium
6	Business rules, etc.	The prediction is based on applicant income, employment, education, family status, housing type, and credit history.	High
7	Compliance to laws or regulations	Protect applicant data and comply with applicable data privacy and banking regulations.	High
8	*Other*	Provide a simple web interface view results.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**
1	Performance & Speed	The system should generate predictions quickly.	Prediction displayed in less than 2 seconds

2	Scalability	The system should support multiple users and increasing application data.	Supports 1,000+ users without performance degradation
3	Security & Data Privacy	Protect user information through secure authentication and encrypted data storage.	Passwords encrypted and data transmitted securely (HTTPS)
4	Reliability & Availability	The application should be available with minimal downtime.	99.9% uptime
5	Usability & Accessibility	The interface should be simple, responsive, and easy to use.	*Accessible on desktop and mobile browsers*
6	Maintainability	The system should be easy to update, maintain, and retrain with new datasets.	Modular Python and Flask code with reusable components